



Expt No 11
INDUCTION MOTOR MACHINE AS GENERATOR AND MOTOR

Aim:

- i) To operate the 3 phase induction machine as (a) induction motor and (b) induction generator.
- ii) To conduct load test in both generating and motor modes.
- iii) To plot efficiency Vs output curves
- iv) To plot output Vs Slip and hence determine the hysteresis power.

Machine details:

Instruments Required :-

S.No	Name of Apparatus	Specifications
1	Voltmeter	
2	Voltmeter	
3	Ammeter	
4	Ammeter	
5	Wattmeter	
6	Wattmeter	

< students are expected to calculate instrument ranges based on the machine ratings before start of experiment and get it approved before connections are made >

Principle

An induction generator is asynchronous in nature because of which it is commonly used in windmill generator since a windmill runs at non-fixed speed. These are used in remote areas to supplement power received from weak transmission links.

If the induction machine is driven at a speed greater than synchronous speed by a prime mover, the direction of induced torque reverses and it acts as an induction generator. The rotating magnetic field is set up by the magnetizing current drawn from the mains. Based on the way with which the generator gets the required lagging reactive power, they are classified into i) separately excited induction generator and ii) self-excited induction generator.

Separately excited induction generator: Induction machine connected to supply mains and driven at super-synchronous speed by its prime mover is called a separately excited induction generator. The generator draws the required lagging reactive power from the mains.

Self excited induction generator : In this generators, the necessary lagging reactive power for its excitation is obtained by a capacitor bank connected across the generator terminals. If the rotor of the machine is driven by its prime-mover, the presence of residual flux causes a small EMF to be induced in the stator windings proportional to the rotor speed. This voltage impresses over the 3-phase supply and causes a rise in leading current drawn from the capacitor bank which is equivalent to lagging current supplied to the generator. The flux set up by this current assists the initial residual flux. Hence the net flux will increase causing the voltage to build up further. The steady state voltage induced on no load is given by the intersection of magnetization characteristic of the machine and capacitance VA characteristic,